
SOLAR POWER PLANTS AND THEIR APPLICATION

ANSYS Fluent CFD Modeling of Solar Air-Heater Thermoaerodynamics

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Abstract—Numerical modeling of thermal and aerodynamic processes taking place in a solar air heater with light-absorbing L-shaped fins is performed. The study is carried out by means of an ANSYS Fluent Solver with an integrated solar calculator for the city of Samara, Russia. The influence of the design (the step between the fins) and technological variables (the Reynolds number) on heat-exchange processes and flow aerodynamics was established during the CFD analysis. Pictorial contours are determined for the velocity distribution, pressure, and other characteristics of the airflow in the solar air-heater box. Thermoaerodynamic characteristics and effective values of technological parameters for the effective use of a solar air heater are studied. The maximum thermoaerodynamic characteristic for the solar air heater with a finned light-absorbing surface is 1.91 for $Re = 1500$, and the minimum test fin pitch is 30 mm.

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INTRODUCTION

Every year, the use of solar energy increases. From 2005 to 2015 the amount of useful solar energy increased from 3.7 to 236.7 TWh [1]. Various devices are used to convert solar radiation into useful energy—photoelectric panels [2–4], solar collectors [5, 6], solar air heaters [7, 8], and so on.

Solar air heaters (SAHs) are widely used in small-scale power industry because of their simplicity of construction, low capital cost, and reliability of equipment. They are used not only for heating, but also for drying agricultural products, heating residential and nonresidential premises, drying wood, etc. In a rectangular channel of a solar air heater, solar radiation, impinging on a dark light-absorbing surface, heats it, and the heated surface transfers the heat to the convected air moving in the channel. The heat-transfer coefficient between the heated light-absorbing surface and the working medium is extremely low in this air heater, which leads to a decrease in the SAH efficiency. To increase the SAH efficiency, standard methods of heat transfer enhancement are used [9]—studding, finning, etc.—which in turn cause increased aerodynamic resistance of the duct.

Researchers have proposed various geometric shapes of the light-absorbing surface to enhance heat transfer. An experimental investigation of SAH thermal performance performed by M.K. Gupta et al. [10], showed that the use of a metal mesh, which creates roughness of the light-absorbing surface, increases the efficiency compared to a smooth surface

by 18–23% for different conditions. Finning has been studied as a technique increasing the heat-transfer coefficient in a solar air heater [11]. These publications show that the creation of artificial roughness on the lower side of a light-absorbing surface contributes to heat-transfer enhancement, but causes frictional losses in the channel. Therefore, the optimal fin and stud configuration for efficient solar air-heater performance needs to be found.

With the development of computational modeling of hydro- and aerodynamic processes (CFD modeling), the search for optimal channel designs and their surfaces has been greatly simplified. A computational analysis of various rough surfaces was carried out in [12] and determined the optimum geometry of an air heater that was studied experimentally after CFD modeling.

This paper presents a numerical study of the thermoaerodynamic performance of a solar air heater the light-absorbing surface of which has modified rectangular L-shaped fins. The objective of the investigation is to study the effect of the relative roughness pitch and air velocity on the heat exchange and flow aerodynamics, as well as to determine the optimal location of L-shaped fins to maximize heat transfer and minimize aerodynamic drag. Using the developed model, technological and design parameters of the solar air heater were determined providing the maximum thermal efficiency. The study was performed using ANSYS Fluent adjoint solver.

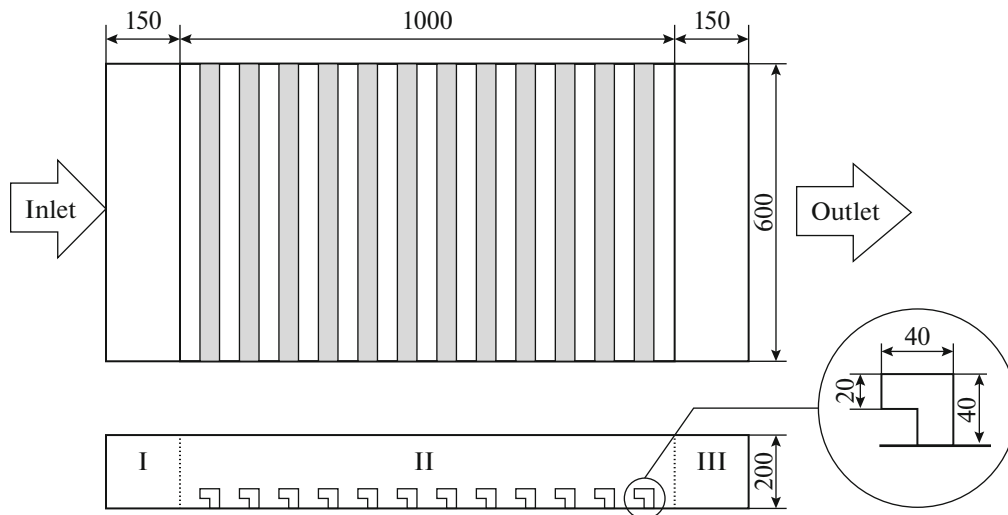


Fig. 1. Computational domain: (I) air-inlet zone, (II) heat-exchange zone, and (III) outlet zone.

CFD MODELING

The problem is solved in a three-dimensional CFD grid. It is assumed that the gravitational and magnetic fields of the Earth are ignored, the airflow is single-phase and incompressible, and the airflow is stable and steady-state. It is also believed that the side shell of the solar air heater is thermally perfect and there is no heat loss through the filler structure. Low density of heat-transfer fluid and its relatively low velocity, which is close to the natural convection velocity, are thermal features of solar air heaters [13]. Because of this, it is possible to use simplified thermal models describing the thermophysical processes of an SAH. However, the used technique makes it possible solve such problems quite accurately without additional assumptions and computing costs. The main stages of the CFD modeling are described in the relevant sections.

COMPUTATIONAL DOMAIN

The geometric characteristics of the computational domain are shown in Fig. 1. The air channel of the solar air heater is divided into three zones: an inlet zone, a heat exchange zone and an outlet zone, respectively. It is believed that the airflow rate in each elementary section of the inlet zone is constant and uniform. The height and width of the fins are 40 mm, and the fin has a distinctive L-shaped appearance. The values of thermophysical air properties are taken from the built-in Fluent Database. The Reynolds number varies in the range from 4000 to 20000. The computation is performed for various fin configurations. Fin pitches of 30, 40, 50, and 60 mm are considered.

To create the design geometric parameters, the ANSYS DesignModeler module was used.

CFD GRID

Creation of a computational grid is performed in the ANSYS Meshing module. To improve the performance quality, a CFD grid was constructed near the walls of the absorbing surface. Using standard “sizing” and “inflation” functions, a wall boundary layer with a thickness of 5 mm was created. A structure of the CFD grid for a 40-mm fin pitch is shown in Fig. 2. The total number of grid elements is 1086806.

A check of the grid quality showed the absence of elements with negative computational mesh area and orthogonality in the range from 0 to 1, which indicates high quality of the grid.

MATHEMATICAL DESCRIPTION

A mathematical description of the model generally has the form of a system of equations consisting of continuity equations, the law of conservation of momentum, and the law of conservation of energy. To simulate the airflow dynamics, the Navier–Stokes equations are used closed by the standard k – ε turbulence model [14, 15].

The continuity equation is:

$$\frac{\partial}{\partial x_i}(\rho u_i) = 0, \quad (1)$$

where ρ is air density, kg/m^3 , and u is velocity, m/s .

The momentum-conservation equation is

$$\begin{aligned} \frac{\partial}{\partial x_i}(\rho u_i u_j) = & -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] \\ & + \frac{\partial}{\partial x_j}(-\rho u_i u_j), \end{aligned} \quad (2)$$

where μ is a dynamic coefficient of viscosity, Pa s .

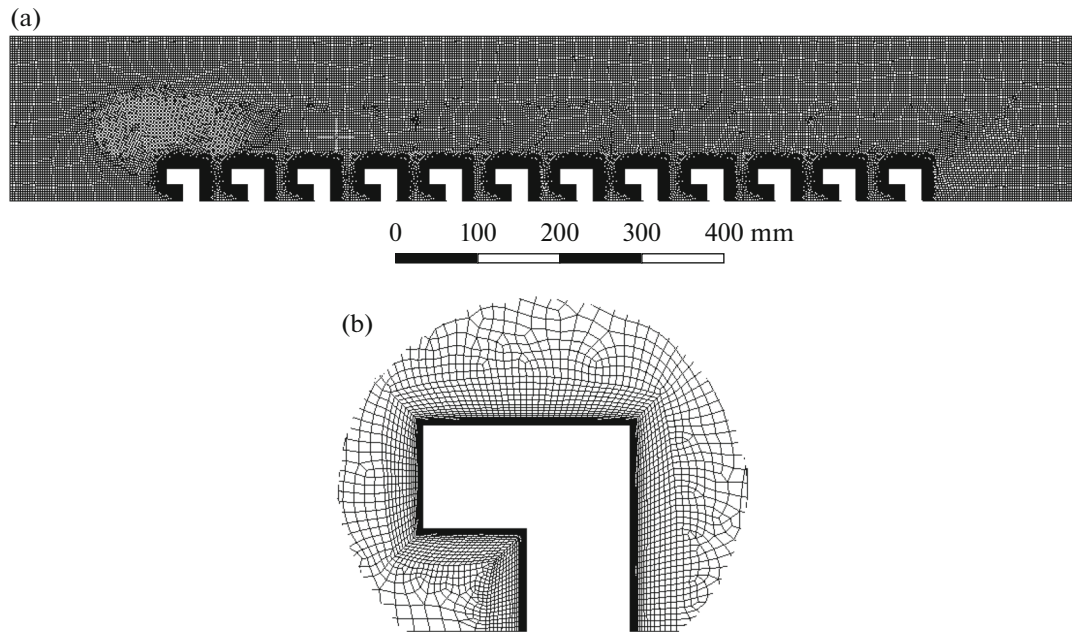


Fig. 2. Structure of the CFD grid: (a) appearance of the CFD grid of the air channel; (b) boundary layer near the fin surface.

The heat-balance equation is

$$\frac{\partial}{\partial x_i} (\rho u_j T) = \frac{\partial}{\partial x_j} \left[(a + a_t) \frac{\partial T}{\partial x_j} \right], \quad (3)$$

where a and a_t are molecular and turbulent thermal diffusivity, respectively, m^2/s , and T is temperature, K.

The initial and boundary temperature conditions are as follows:

$$T_{\text{in}}(t=0) = T_{\text{la}}(t=0) = T_{\text{g}}(t=0) = T_{\text{e}}(t=0), \quad (4)$$

$$\begin{aligned} \frac{\partial T_{\text{in}}}{\partial x} \Big|_{x=0} &= 0, \quad \frac{\partial T_{\text{g}}}{\partial x} \Big|_{x=H} = 0, \quad \frac{\partial T_{\text{la}}}{\partial x} \Big|_{x=0} = 0, \\ \frac{\partial T_{\text{la}}}{\partial x} \Big|_{x=H} &= 0, \end{aligned} \quad (5)$$

where T_{in} , T_{la} , T_{g} , and T_{e} are air temperatures at the inlet to the heater, light-absorbing surface, glass, and environment, respectively, K.

FLUENT SOLVER SETTINGS

The problem has a steady state solution; the solver is pressure-based. To determine the heat transfer between the light-absorbing surface and the air, the heat balance equation (Energy-on) is used. The Navier–Stokes equation is closed by the standard $k-\epsilon$ model with improved wall functions to improve the quality of boundary-layer computation (Fig. 3).

The quantity of solar radiation is calculated using the ANSYS Solar Calculator, which allows obtaining the intensity of solar radiation depending on the following parameters [16]:

- (a) geographical coordinates;
- (b) date and time of the simulation exercise;
- (c) level angle of the solar air heater to the ground surface; and
- (d) atmospheric transmission factor (sunshine factor).

To solve the system of differential equations describing heat- and mass-transfer processes, a second-order upwind scheme is used. A convergence criterion is chosen on the basis of considerations of minimizing computational calculations and amounted to 10^{-3} . The number of iterations is 1000. The convergence of the solution for various conditions is achieved in the range from 700 to 900 iterations.

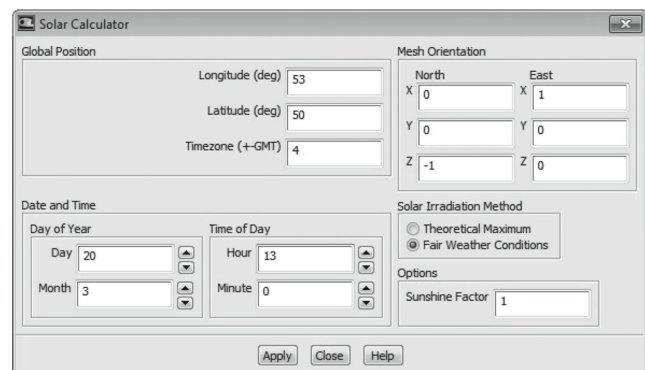


Fig. 3. Solar-calculator home screen.

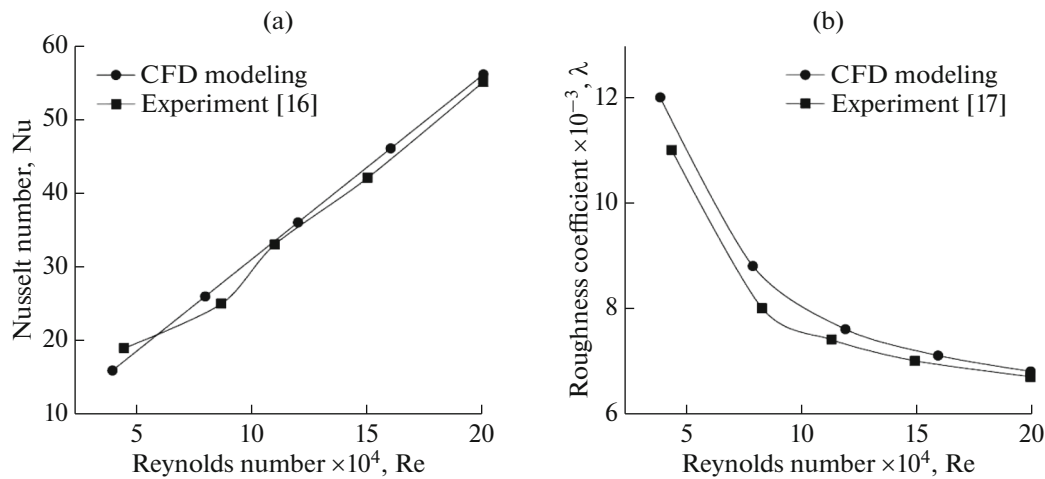


Fig. 4. Comparison of experimental data and the results of CFD modeling for the smooth light-absorbing surface (unfinned).

RESULTS AND DISCUSSION

To verify the results obtained using the CFD model, thermal and aerodynamic characteristics of a smooth (unfinned) surface were compared with the results of a physical experiment performed by Fox et al. [17]. Figure 4a shows the dependence of the Nusselt number on the Reynolds number for the data obtained by the CFD model described in point 2 of this paper and the results obtained in the physical experiment [17]. Also for model verification, the dependences obtained analytically [18] were compared with data obtained from the developed model. From the presented dependences, a good convergence is evident of the data obtained by computer simulation with the results of the physical experiment and the analytical solution, which indicates the correctly chosen settings of the Fluent Solver, such as the turbulence model, solution scheme, convergence criterion, etc.

In general, the adequacy of the developed computer SAH model to a real solar air heater was shown: the behavior of the model quite accurately coincides with the behavior of the simulated object under the same simulation conditions, and the model is convincingly presented with respect to the SAH properties predicted by the model. The paper considers the average Reynolds numbers corresponding to the airflow in the box of the solar air heater, dimensions of which are shown in Fig. 1: $Re = 400, 600, 800, 1200$, and 1600 . The $k-\epsilon$ turbulence model is used in the numerical study, which yields equally accurate results for a wide range of Reynolds numbers [20].

AERODYNAMICS

Figure 5 shows a pictorial airflow-velocity contour for the finned surface, a schematic view of which is shown in Fig. 1. The contour shows the distribution of air velocity in the channel cross section for the Reyn-

olds number $Re = 1200$ for a light-absorbing surface with a 40-mm fin pitch. Similar contours were obtained for the remaining fin pitches in question—30, 50, and 60 mm. For all the surfaces under study, an L-shaped fin creates additional vortices that enhance the heat exchange between the surface and the heated air.

The nature of the vortices formed behind the first five fins located along the airflow is shown in Fig. 6. Due to a frontal airflow impact, a separate flow is formed behind the first fin, which does not have a significant effect on the heat transfer. Further on, the flow stabilizes and stable vortices are formed in the interfin region, full development of which occurs after the passage of the fourth fin.

The presence of L-shaped fins on the light-absorbing surface destroys the boundary layer. Therefore, an increase in the turbulence kinetic energy is observed. A turbulence kinetic-energy contour obtained with the CFD model is shown in Fig. 7. The turbulence kinetic energy is a turbulence force in the airflow field.

The kinetic-energy contour makes it possible to understand the effect of the artificially created surface roughness by L-shaped fins on the intensity of heat exchange between the surface and air. The contour in Fig. 7 shows that the maximum value of the kinetic energy is observed near the light-absorbing surface between the fins, and then it decreases with an increase in the distance from the surface. Thus, the presence of L-shaped fins significantly enhances heat-exchange processes near the light-absorbing surface.

HEAT EXCHANGE

The main purpose of L-shaped fins on a light-absorbing surface is to enhance the heat exchange between the surface and the airflow. The contours in Figs. 5–7 show the factors of gas-flow turbulence. Figure 8 shows the dependence of the Nusselt number

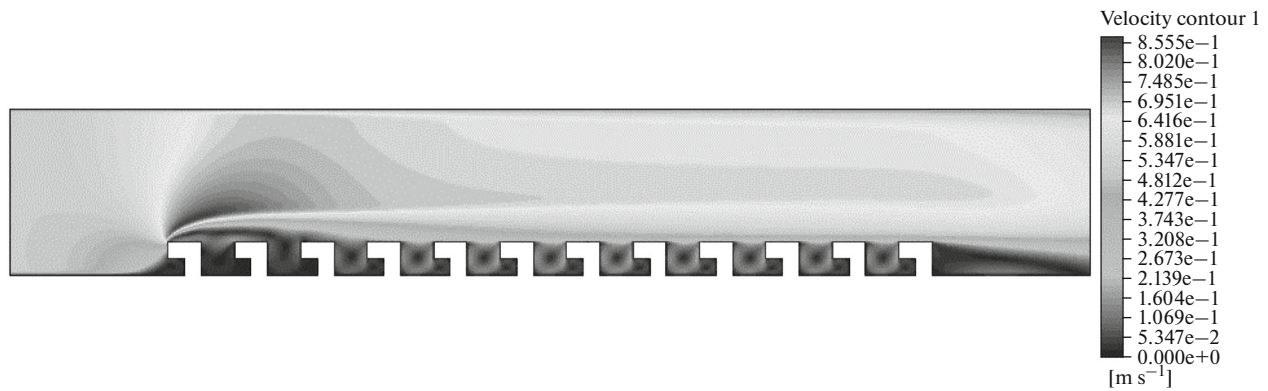


Fig. 5. Airflow-velocity contour for the surface with a 40-mm fin pitch for Reynolds number $Re = 1200$.

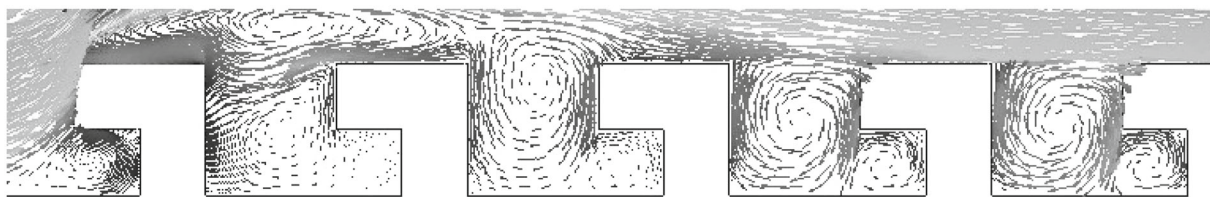


Fig. 6. Vector field of the airflow velocity for the first five fins with a pitch of 40 mm along the airflow, $Re = 1200$.

on the Reynolds number for light-absorbing surfaces with different fin pitches obtained by the developed model for various Reynolds numbers $400 > Re > 2000$.

It can be seen from Fig. 8 that the presence of fins significantly increases the heat exchange between the light-absorbing surface and the airflow in comparison with a smooth (unfinned) surface. Thus, for a 40-mm fin pitch, the Nusselt number increases by more than three times compared to a smooth surface. Figure 8 also shows that the average Nusselt number increases with the increase of the fin pitch. This is because the increase in the distance between the fins leads to a decrease in the kinetic energy of the turbulent flow.

Figure 9 shows the dependence of the air temperature drop at the inlet and outlet of the solar heater for various shapes of the light-absorbing surface during the day. As initial data for the analysis of the influence of these parameters on the air heating value, the following values were used: date March 20, 2017; Samara (GMT + 4); absolutely clear sky (sunshine factor = 1); $Re = 800$; level angle of the solar air heater to the ground surface $\alpha = 60^\circ$.

From Fig. 9 it can be seen that the maximum air heating is observed in the interval between 1 and 2 pm. This is because the angle cosine between the sunbeam and the SAH surface at this time has the maximum value for the middle of the spring in Samara. The maximum air heating value of 25.4°C is achieved for the



Fig. 7. Contour of the air-flow-turbulence kinetic energy for the surface with a fin pitch of 40 mm, $Re = 1200$.

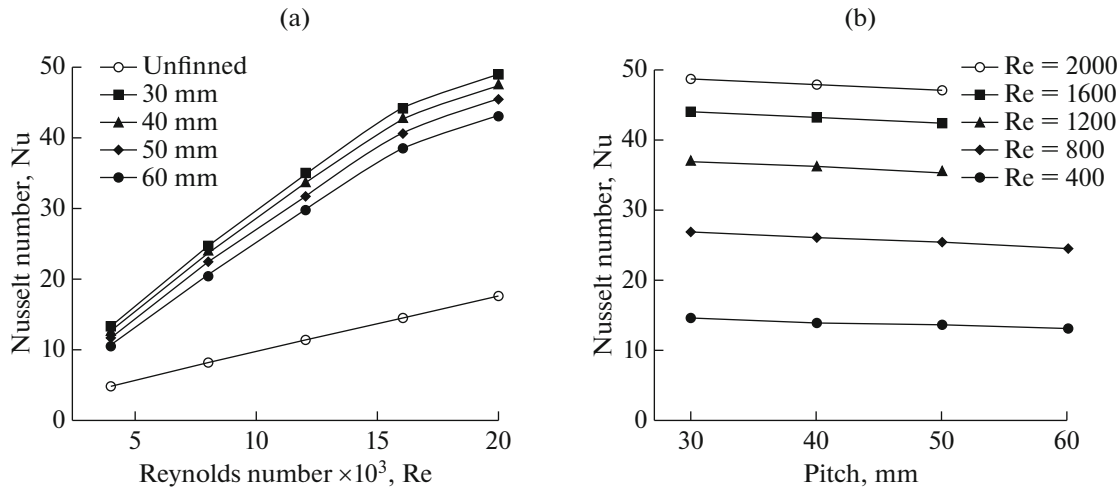


Fig. 8. Variation of the Nusselt number with the Reynolds number for various fin pitches.

minimum inlet air velocity and the minimum 30-mm test pitch.

ROUGHNESS COEFFICIENT

One of the main parameters taking into account the efficiency of using the finned light-absorbing surface is a roughness coefficient. Figure 10 shows a graph of variance of the friction coefficient for a finned light-absorbing surface having a different pitch and for a smooth surface. The friction coefficient in the present CFD experiment is determined from the Darcy–Weisbach formula [19, 20]:

$$\lambda = \frac{\Delta P}{\frac{L}{d_{eqv}} \rho \frac{u^2}{2}}, \quad (6)$$

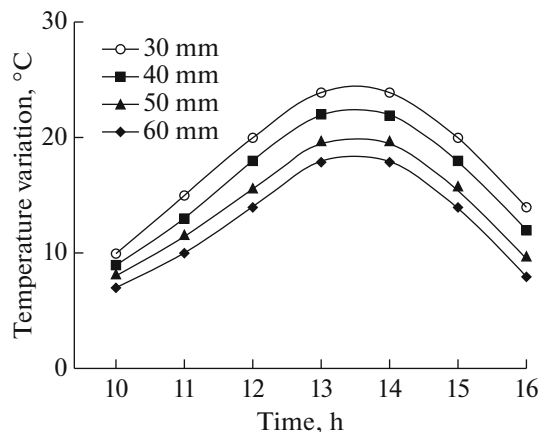


Fig. 9. Air-temperature variation during the day for different light-absorbing surfaces.

where ΔP is pressure losses in SAH, Pa; L is a characteristic length, m; d_{eqv} is an equivalent diameter, m; ρ is air density, kg/m^3 ; and u is air velocity, m/s.

According to Fig. 10, the L-shaped fin setting on a light-absorbing surface of a solar heater creates an additional roughness that destroys the laminar wall layer. This leads to the appearance of extra aerodynamic resistance and, as a result, to a significant increase in the roughness coefficient in comparison with a smooth surface. So for $\text{Re} = 8000$, the roughness coefficient is 3.8–4.2 times higher for different fin pitches than for a smooth (unfinned) surface. An increase in the Reynolds number causes suppression of the laminar sublayer and, consequently, the value of the roughness coefficient decreases with an increase in the Reynolds number, as shown in Fig. 10b. As discussed above, the increase in the Reynolds number increases the turbulence intensity and, as a result, the heat exchange grows with a smaller value of the friction coefficient.

A pressure contour obtained with the CFD analysis for $\text{Re} = 1200$ and a 40-mm fin pitch is shown in Fig. 11.

Figure 11 shows that the maximum aerodynamic resistance along the airflow is offered by a fin. Minimization of this resistance is possible due to the use of a first fin of another shape, for example, an L-shaped fin, inverted in the airflow direction.

THERMOAERODYNAMIC CHARACTERISTIC

It was found in the CFD study that was carried out that the artificial roughness in the form of additional fins on a light-absorbing surface greatly enhances the heat exchange between the wall and the heated air. Nevertheless, the additional roughness leads to higher aerodynamic resistance and, as a consequence, to an increase in the energy consumption for air pumping through the SAH air channel. Therefore, for an objec-

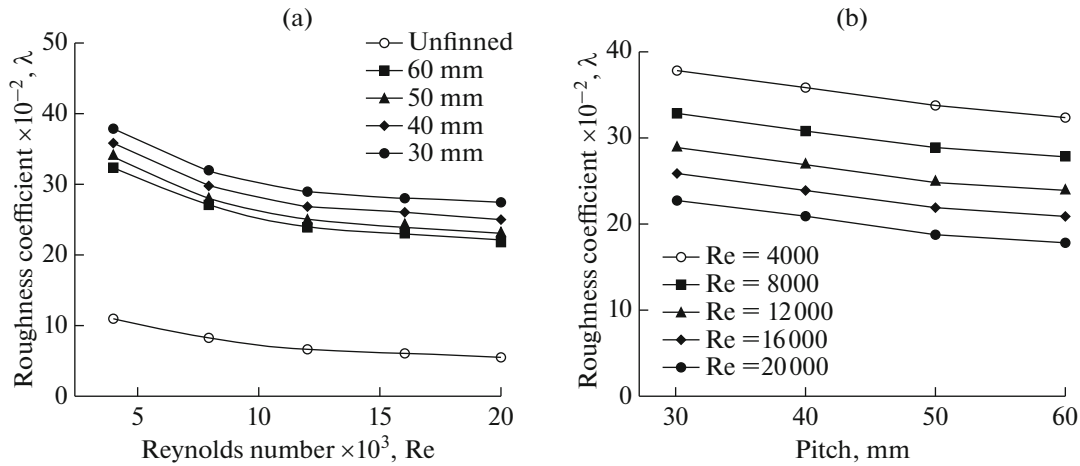


Fig. 10. Influence of Reynolds number on roughness coefficient for various fin-pitch values.

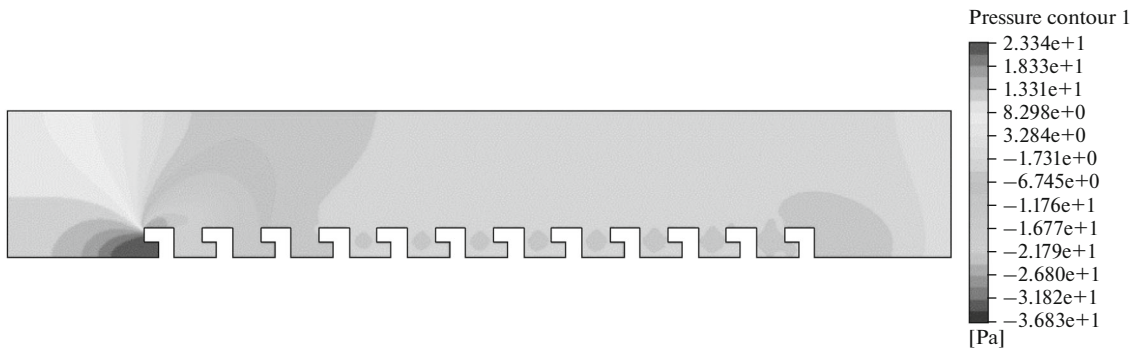


Fig. 11. Pressure contour in the air channel of the solar air heater for $Re = 1200$ and a 40-mm fin pitch.

tive analysis, it is necessary to evaluate a generalized thermoaerodynamic characteristic of the simulated air heater, and not only thermal and aerodynamic ones.

A characteristic of the system based on the interrelation between thermal and aerodynamic parameters is called a “thermoaerodynamic” characteristic, for which quantitative evaluation of which the following expression is used [21]:

$$\eta = \frac{Nu/Nu_{uf}}{(f/f_{uf})^{1/3}}. \quad (7)$$

The change in the thermoaerodynamic characteristic for various fin pitches and Reynolds-number variables is shown in Fig. 12.

From the dependences in Fig. 12 obtained in the numerical simulation, it was established that the value of the thermoaerodynamic characteristic varies between 1.62 and 1.91 in the range of the parameters under study. It is apparent from Fig. 12 that the thermoaerodynamic characteristic has a maximum value for the 30-mm fin pitch at Reynolds number $Re = 1500$. The minimum value of the thermoaerodynamic characteristic is observed for the 60-mm fin pitch. Fig. 12 also shows that an increase in Reynolds number $Re > 1500$ leads to a significant decrease in the thermoaerodynamic characteristic. This is due to the fact that an

increase in the Reynolds number is usually associated with an increase in airflow velocity, which has a significant effect on the roughness coefficient in accordance with relationship (7).

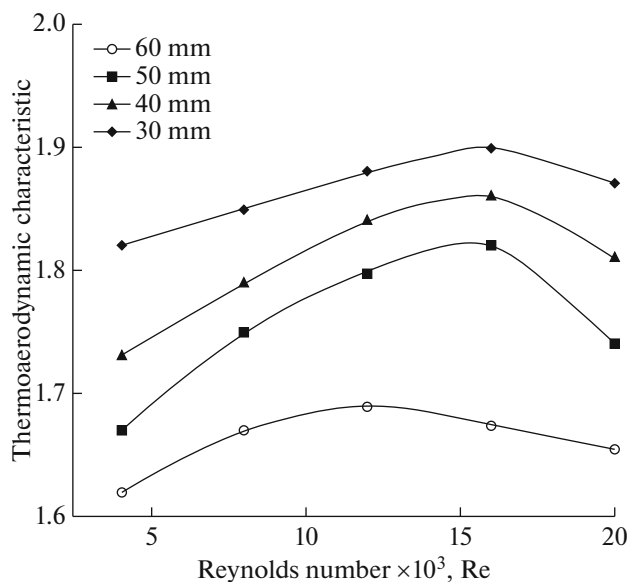


Fig. 12. Dependence of the thermoaerodynamic characteristic on the Reynolds number for various fin pitches.

CONCLUSIONS

Numerical modeling was performed of the processes taking place in a solar air heater the light-absorbing surface of which has an artificial L-shaped finned roughness. The simulation was carried out using an ANSYS Fluent Solver with an integrated Solar calculator. The influence of the design (fin pitch) and technological variables (the Reynolds number) on the heat-exchange and flow aerodynamic processes was established in the CFD analysis. The velocity distribution, pressure, contours and other characteristics of the airflow in the solar air-heater box were obtained.

It was established that the average Nusselt number increases with a decrease of the fin pitch. On average, for various values of the Reynolds number, the Nusselt number for the 30-mm pitch is 5–7% greater than the Nusselt number obtained for the 60-mm pitch.

The presence of an artificial roughness on the light-absorbing surface leads to an increase in aerodynamic resistance and, as a consequence, the roughness coefficient. The maximum aerodynamic resistance is presented by the first fin along the airflow. The value of the roughness coefficient increases with the decrease of the fin pitch. The difference between the roughness coefficient for the 30- and 60 mm pitches is 7–10%. The maximum increase in the roughness coefficient compared to the smooth surface is 3.46 for $Re = 4000$ with a fin pitch equal to 30 mm. The increase in the Reynolds number causes suppression of the laminar sublayer, and, consequently, the roughness-coefficient value decreases with the increase in the Reynolds number, which agrees with classical fluid dynamics.

The value of the thermoaerodynamic characteristic used to determine the optimum SAH design and process parameters (air velocity), which is in the range between 1.62 and 1.91, was determined for various fin pitches and Reynolds numbers. The maximum value of the thermoaerodynamic characteristic for the solar air heater with a finned light-absorbing surface, which is optimal for these parameters, is 1.91 for $Re = 1500$ and a minimum test fin pitch of 30 mm.

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